Pediatric Anaphylaxis / Allergic Reaction

PM-01 | Pediatric Medical

Quick flow

01 Suspected Allergic Reaction [EMT / AEMT / Paramedic]

Assess airway, breathing, circulation, exposure, onset, skin, GI, and cardiovascular findings. Remove the trigger when safe.

02 Anaphylaxis Suspected? [EMT / AEMT / Paramedic]

Airway/breathing compromise, hypotension/poor perfusion, or multi-system symptoms after likely allergen: treat immediately. Do not wait for every criterion when concern is high.

03 IM Epinephrine Now [EMT / AEMT / Paramedic]

Epinephrine 1 mg/mL: 0.01 mg/kg IM in the mid-outer thigh; maximum 0.3 mg prepubertal child / 0.5 mg adolescent. Repeat every 5 minutes as needed.

04 Support Airway + Breathing [EMT / AEMT / Paramedic]

Oxygen/ventilation as indicated; albuterol only for lower-airway bronchospasm. Prepare early for rapid airway deterioration.

05 Anaphylactic Shock? [AEMT / Paramedic]

Normal saline 20 mL/kg IV/IO with reassessment after each bolus. Repeat IM epinephrine and contact Medical Control early for refractory symptoms.

06 Reassess + Transport [EMT / AEMT / Paramedic]

Continuous monitoring, repeat epinephrine when indicated, and ED evaluation for every child treated with epinephrine.

Overview

- Epinephrine is the first-line treatment for pediatric anaphylaxis and must not be delayed for vascular access, antihistamines, albuterol, or corticosteroids.
- Treat suspected anaphylaxis promptly when airway/breathing compromise, hypotension/poor perfusion, or multi-system symptoms follow a likely allergen.

Provider-Level Actions

Use a length- or weight-based pediatric system and reassess after every intervention.

EMT

- Remove the trigger when safe, support airway/ventilation, give IM epinephrine for suspected anaphylaxis, and apply cardiac/SpO_2 monitoring when available.
- For mild isolated skin symptoms, provide supportive care and reassessment. Do not use oral medication for anaphylaxis or an unprotected airway.

AEMT

- Perform all EMT care plus IV/IO access, normal saline for anaphylactic shock with reassessment after each bolus, nebulized albuterol for bronchospasm, and authorized repeat IM epinephrine.
- Do not delay epinephrine, airway support, or transport for antihistamine administration.

Paramedic

- Perform all prior care plus advanced airway management, Paramedic-only diphenhydramine for isolated or persistent cutaneous symptoms after stabilization, and Medical-Control-directed epinephrine infusion for refractory anaphylaxis after repeated IM doses.

- Give methylprednisolone 2 mg/kg IV (maximum 125 mg) only as an adjunct after epinephrine. Do not use a routine IV/IO epinephrine bolus for refractory anaphylaxis.

Indications

- Pediatric allergic reaction or suspected anaphylaxis after a likely exposure.

Treatment steps

- IM epinephrine is the time-critical definitive treatment. Do not delay it for IV/IO access, diphenhydramine, corticosteroids, nebulizers, or transport preparation.
- Give albuterol only for lower-airway bronchospasm. It does not reverse upper-airway edema, hypotension, or shock.
- For anaphylactic shock, give normal saline 20 mL/kg with reassessment after each bolus; repeat IM epinephrine and open PM-03 for persistent shock.
- All children treated with epinephrine for anaphylaxis require ED evaluation and monitoring for recurrence, even if symptoms resolve.

Medications

Epinephrine 1 mg/mL (1:1,000)

0.01 mg/kg IM in the mid-outer thigh; maximum 0.3 mg prepubertal child / 0.5 mg adolescent; repeat every 5 minutes as needed

Albuterol

Under 20 kg: 2.5 mg nebulized; 20 kg or greater: 5 mg nebulized for lower-airway bronchospasm

- Adjunct only; does not treat upper-airway edema, hypotension, or anaphylaxis itself.

Diphenhydramine

Paramedic: 1 mg/kg PO/IV/IO/IM; maximum 50 mg

- Adjunct for isolated or persistent cutaneous symptoms only after stabilization.

Normal saline

20 mL/kg IV/IO for anaphylactic shock; reassess perfusion and lungs after each bolus

Methylprednisolone

2 mg/kg IV; maximum 125 mg

- Adjunct only; do not delay or replace epinephrine.

Epinephrine infusion

Medical-Control-directed for refractory anaphylaxis after repeated IM epinephrine; do not use a routine IV/IO epinephrine bolus

Warnings / precautions

- Do not delay IM epinephrine for IV access, diphenhydramine, corticosteroids, or albuterol.
- Do not give oral medication to a child with decreased mental status, inability to swallow, or an unprotected airway.
- Do not use intranasal epinephrine as the EMS anaphylaxis treatment pathway.
- Do not use a routine IV/IO epinephrine bolus for refractory anaphylaxis.

Clinical pearls

- A falling blood pressure may be late; poor perfusion and respiratory compromise require action before hypotension develops.
- The lateral thigh is the preferred IM epinephrine site for reliable absorption.

Related pathways / contacts

AR-05 Pediatric Airway - Open for pediatric BVM and airway support.

AR-06 Pediatric Failed Airway - Open for rapidly worsening or ineffective airway management.

PM-03 Pediatric Hypotension / Shock - Open for persistent shock after epinephrine and fluid reassessment.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS PM-01 Pediatric Anaphylaxis / Allergic Reaction protocol, approved October 26, 2026. | Anaphylaxis: A 2023 practice parameter update (AAAAI/ACAAI). | American Academy of Pediatrics: Epinephrine for First-aid Management of Anaphylaxis.

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